

Load Interconnection System Impact Study Report

For Queue Project Q11-0079

Luka Tap 138 kV
December 01, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0079. The Interconnection Customer proposes a 42.1 MW large load (data center) facility interconnecting at the Luka Tap 138 kV substation in the Tennessee area. The request was received on October 26, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 69 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$58,414,000.

3. Network Impacts

3.1 Load Deliverability Analysis

The Beardsley St to Red Hills 138.0 kV line is overloaded to 131.9% of its 234 MVA emergency rating for the loss of the Bear to Clover 345.0 kV. The project contributes 5.0% DFAX (2.1 MW) to this overload.

For the loss of Clover to Bear 345.0 kV, the Vaseaux Lake to Red Hills 138.0 kV line loads to 124.8% of its 239 MVA rating. Project DFAX: 5.0%.

For the loss of Clover to Bear 345.0 kV, the McClain Energy Facility Substation to Vaseaux Lake 138.0 kV loads to 123.1% of its 240 MVA rating. Project DFAX: 5.0%.

The Aspen 345-138 kV Auto is overloaded to 115.2% of its 307 MVA emergency rating for the loss of the Bath County to Longview2 345.0 kV line. The project contributes 7.4% DFAX (3.1 MW) to this overload.

The Vail to N Bristol 138.0 kV is overloaded to 102.5% of its 264 MVA emergency rating for the loss of the Spring Crk to Sullivan Gardens 345.0 kV. The project contributes 49.8% DFAX (21.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beardsley St to Red Hills 138.0 kV	Rebuild line with higher-capacity conductor	\$7,910,000	100% (\$7,910,000)
Vaseaux Lake to Red Hills 138.0 kV line	Rebuild line with higher-capacity conductor	\$24,070,000	100% (\$24,070,000)
McClain Energy Facility Substation to Vaseaux Lake 138.0 kV	Rebuild line with higher-capacity conductor	\$10,095,000	100% (\$10,095,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aspen 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$14,349,000	100% (\$14,349,000)
Vail to N Bristol 138.0 kV	Reconductor line	\$1,990,000	100% (\$1,990,000)
	Total allocated to project		\$58,414,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.